

Chapter 13 — Documentation and metadata

Standards require concrete records

Learning objectives

- By the end of this part

Research documentation demands

- Beyond the documentation components listed in Chapter 8
- The goal is a checkable published path, not only an internal ops record

Example 13.9 — Undocumented Methodology Blocks Replication

- Example 13.9 — hands-on module
- Example 13.9 shows the cost of unclear methods
- In a medical study
- Surveys of researchers have long found that inadequate documentation of methods and data
- Explore the chapter example module
- View files: ``modules/chapter13/example9/``

Example 13.10 — Repository Metadata Enables Discovery

- Example 13.10 — hands-on module
- Metadata types and roles are covered in Chapter 8
- What matters for open science is publication-grade metadata
- License, so related searches can discover it beyond the original study
- Explore the chapter example module
- View files: ``modules/chapter13/example10/``

Example 13.11 — Unpinned Dataset Versions Break Replication

- Example 13.11 — hands-on module
- Versioning tools are developed in Chapters 8 and 12
- The exact dataset version used in a published analysis must be identifiable and
- Example 13.11 shows what breaks without pinned versions
- Explore the chapter example module
- View files: ``modules/chapter13/example11/``

Takeaways

- Undocumented methods leave published findings checkable only in name
- Publication-grade metadata turns repositories into discovery systems, not dump sites
- Pinning the exact dataset version used in a paper is required for faithful replication
- Chapter 8 and Chapter 12 supply the operational toolkit

Next

- Complete the quiz for this part
- The next part introduces FAIR principles, Findable, Accessible, Interoperable